CSCI 373 SENIOR RESEARCH SEMINAR · SPRING 2025

SHAP explains model predictions, with limits

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[CSB and SJU logo, from the official file]

MOTIVATION

Decisions need reasons

Models now help decide loans, hiring and medical care. The EU AI Act and GDPR expect meaningful information about automated decisions. To find bias and errors, we must see why a model predicted what it did.

THE ONE THING TO REMEMBER

SHAP exposed a pattern my earlier analysis missed, yet its math can still mislabel which features matter. Pair it with other methods.

EXPLORATION

XGBoost plus TreeSHAP

I trained XGBoost to predict survival for 887 Titanic passengers, then explained it with SHAP in a Jupyter notebook. SHAP ranked the features:

1 Sex · 2 Class · 3 Age · 4 Fare

THESIS

Useful, but not sufficient

SHAP is a valuable explanation tool, but its Shapley-based attributions can diverge from logical relevance, so it should be used with complementary methods.

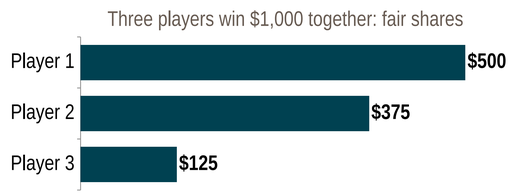
SCOPE

TreeSHAP applied to an XGBoost survival model on the Titanic dataset, with a review of the fairness axioms and their limits.

FOUNDATIONS

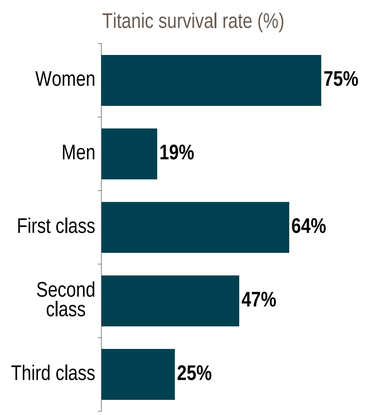
Shapley values split the credit

Each feature is a player in a cooperative game. Its Shapley value is its average marginal contribution across every coalition.



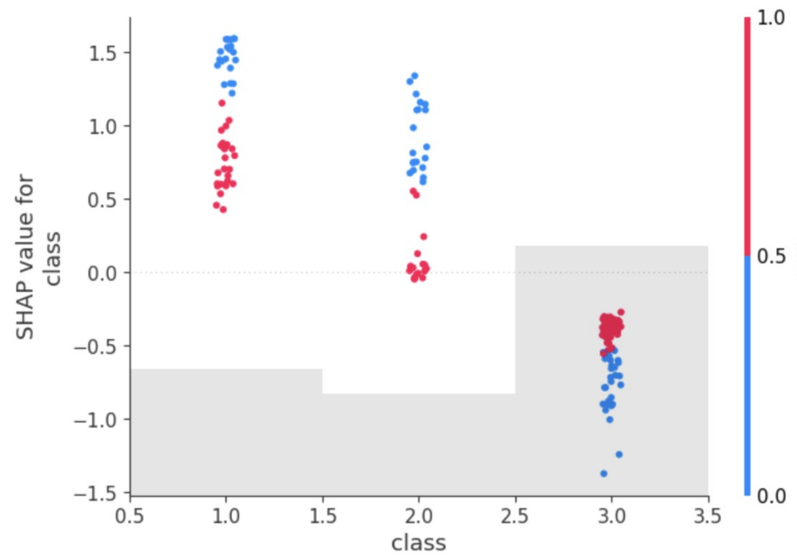
TreeSHAP computes these values exactly for tree models.

What the lab analysis showed



First-class women: about 95%. Third-class men: under 20%.

What SHAP added: class matters more for women



Each dot is one passenger. Blue: women. Red: men. Figure 11 of the paper.

WHAT THE FIGURES SHOW

The lab analysis (left) gave survival rates by sex and by class. SHAP values from the XGBoost model (right) show class moving predictions more for women: first and second class raised their odds more, and third class lowered them more than it did for men.

LIMITS

Where SHAP can mislead

Huang and Marques-Silva (2023) show that Shapley values can:

1. Score an irrelevant feature above zero
1. Rank an irrelevant feature over a relevant one
1. Score a relevant feature at zero

Mathematical impact is not logical relevance.

FUTURES

More rules, more built-in tools

Within a year, SHAP-type tools grow as the EU AI Act and GDPR push transparency, while explanation moves inside the models themselves.

GO DEEPER

Scan for the full paper

Full State-of-the-Field paper
[90-second video]
[Notebook and code]
Or type: [short link]

Sources. Lundberg, Erion and Lee (2019). Huang and Marques-Silva (2023). Full list in the paper.

AI acknowledgment. [Which tools you used and for what.]

Thanks. [Instructor, reviewers and others who shaped the work.]